Name: Owen Nda Diche **Student ID: 24152506**

What Does an Autoencoder Remember?

Investigating Latent Dimension and Quality of Reconstruction.

Technique	Fully connected autoencoder	**Focus**	Effect of latent dimension
Dataset	Fashion-MNIST	**Latent dimensions**	2, 8, 16, 32, 64
GitHub Repository: https://github.com/owendiche-sys/Autoencoder-Latent-Dimension-Tutorial			

Objective

This tutorial describes how an autoencoder can use the size of the so-called bottleneck, or more generally the latent dimension, to determine what a given image can preserve during compression. This is not merely to train a model, but to demonstrate why the bottleneck size is an important consideration and how a practitioner can make a more informed choice.

Introduction

Autoencoders are neural networks that are expected to recreate their input. They accomplish this by training two related processes encoder which maps the input data to a low-dimensional data representation and decoder which recovers the original input based on the low-dimensional data representation. The condensed space is known as the latent space or bottleneck. Dimensionality reduction, feature learning, denoising and anomaly detection Autoencoders are also common in tasks that do not need class labels to learn a compact and useful representation of data (Hinton and Salakhutdinov, 2006; Goodfellow, Bengio and Courville, 2016).

This tutorial focuses on one important design choice in autoencoders: latent dimension. The latent dimension determines the amount of information that the bottleneck is able to hold. When it is smaller, then the model can be deprived of necessary structure. When larger it might result in higher reconstruction quality at the expense of weakened compression. This tutorial will demonstrate the sensitivity of latent dimension to the quality of reconstruction of a simple autoencoder trained on Fashion-MNIST (Xiao, Rasul and Vollgraf, 2017).

The importance of latent dimension.

The main principle of an autoencer is compression. This causes the model to filter the input by a bottleneck triggering it to memorize the most useful information and forget unhelpful detail. The size of this bottleneck in practice is very important in determining what the model can learn. A very narrow latent space can only be used to record coarse structure whereas a larger size can be used to record finer visual detail. Due to this reason, one of the most significant hyperparameters in designing an autoencer is latent dimension (Goodfellow, Bengio and Courville, 2016).

The tutorial poses a rather straightforward question: in how little time can a fully connected autoencoder compressed information before its ability to reconstruct starts to fail?

Dataset

It is the experiment based on Fashion-MNIST, which is a dataset consisting of 28 x 28 grayscale images of clothing (shirts, shoes, trousers and coats) (Xiao, Rasul and Vollgraf, 2017). Fashion-MNIST has been selected due to its ease of use and interpretability through visual aids, as well as being informative enough, albeit marginally, than the handwritten digits, in displaying the difference in reconstruction. Autoencoders are not trained with class labels, as the objective of the autoencoder is to recreate the original input image and not to make a prediction.

Method

The implementation was of a simple fully connected autoencer in PyTorch (PyTorch Documentation, n.d.). All the input images were reduced to a 28 x 28 pixel image which was then flattened into a 784 length vector. This input was reduced by the encoder into 256 and 128 hidden layers, and then it was mapped to the latent space. The decoder was then used to reconstruct the image by inverting this process so as to come up with 784 outputs and then by applying the sigmoid activation to ensure that the values of every pixel remained between 0 and 1. It was the latent dimension that altered in experiments as the only aspect of the architecture. Five values were tested: 2, 8, 16, 32 and 64. This produced the extreme compression to less constraint compression.

Mean squared error (MSE) was used as the reconstruction loss, and Adam optimiser was used to train the model. The Fashion-MNIST was split into a sub-set to train 15 epochs to ensure the notebook remained feasible and reproducible. Final test reconstruction loss, training curves, reconstructed image examples and a 2D latent-space visualisation were used as methods to measure performance with a 2 latent-dimension model.

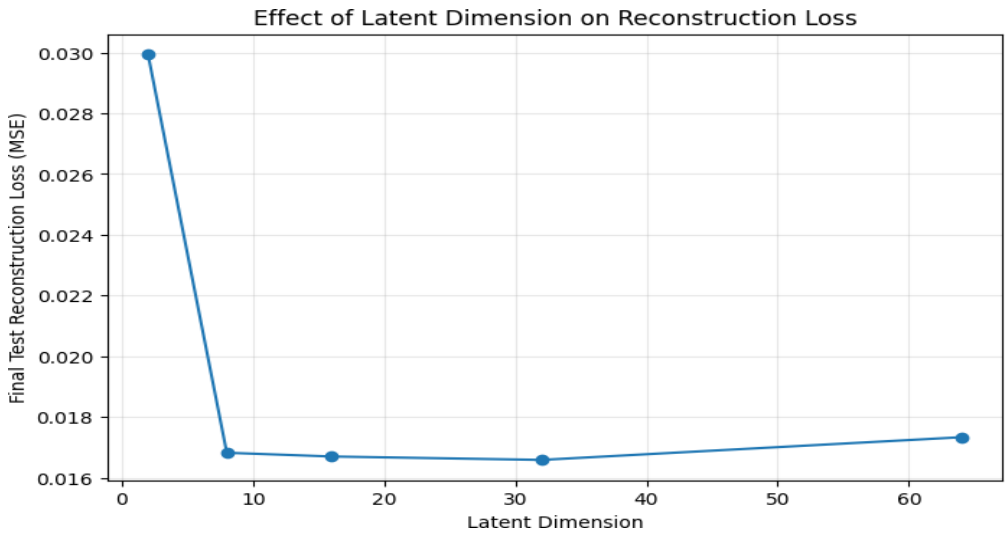

Figure 1. Loss on final test reconstruction by various latent dimensions. The sudden change between the latent dimension 2 and the larger value is an indication that there is a significant information loss occurring due to extremely small bottlenecks.

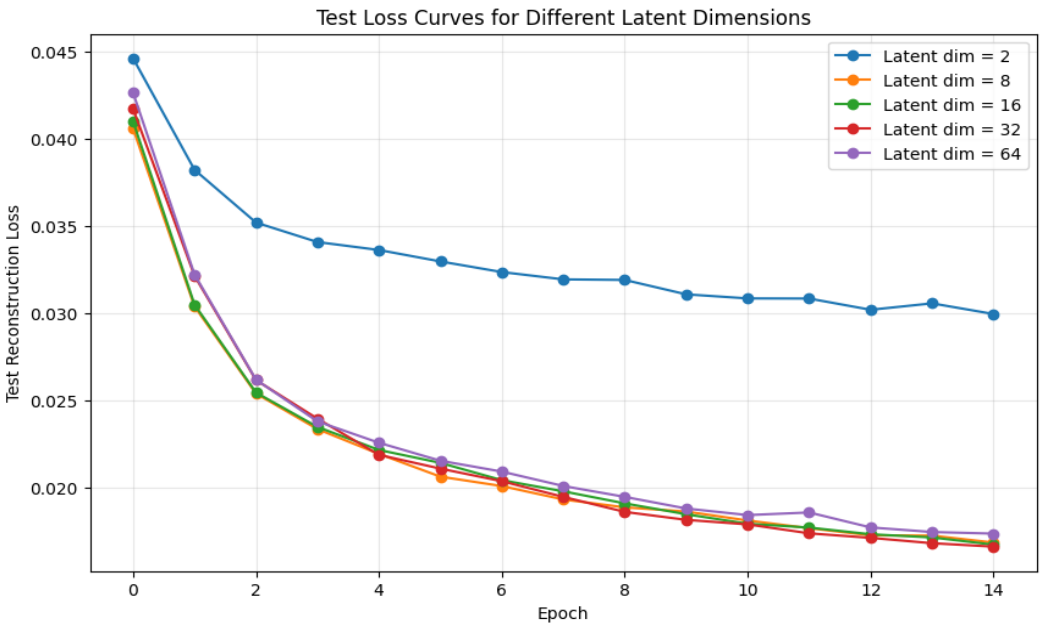

Figure 2. Loss curves of 15 epochs. Latent dimension 2 is also obviously worse than all other models with latent dimension 8, 16, 32 and 64 all converging to the same lower losses.

Results

The findings indicated that there was a significant correlation between latent dimension and reconstruction quality. The latent 2 model was significantly worse than all the other models. Its loss was significantly bigger in its final test reconstruction and the reconstructed images were more blurred. It means that it is too limiting to reduce an image of Fashion-MNIST to two numbers to capture any significant visual information.

Latent Dimension	Final Test Loss
2	0.029938
8	0.016821
16	0.016698
32	0.016586
64	0.017333

Table 1. Loss in reconstruction of final test on each dimension of latent. The least loss was 32 at latent dimension whereas latent dimension 2 did so much more poorly than the rest.

With a latent dimension increased by contrast to 8, 16, 32 or 64, both test loss and reconstructions were greatly reduced and visually improved. The greatest increase was seen between the latent dimensions 2 and 8, indicating that the biggest penalty is offered by over-compression. Thereafter the disparities fell well below that. The results of 32 in this experiment were the lowest final test loss, but that of 8, 16, 32 and 64 were quite similar. This implies that when the bottleneck size is large enough to encode the major structure of the clothing items, an increase in it does not bring significant returns.

This interpretation was assisted by the reconstruction grid. At latent dimension 2, lines were retained loosely and a number of pieces of clothing looked washed or simplified. The models at bigger latent dimensions had more distinct silhouettes and contrast. But the finest reconstructions were distorted somewhat. This is not surprising, as the tutorial employs a fully connected autoencoder, which is not so effective at preserving the spatial image structure compared to a convolutional autoencoder.

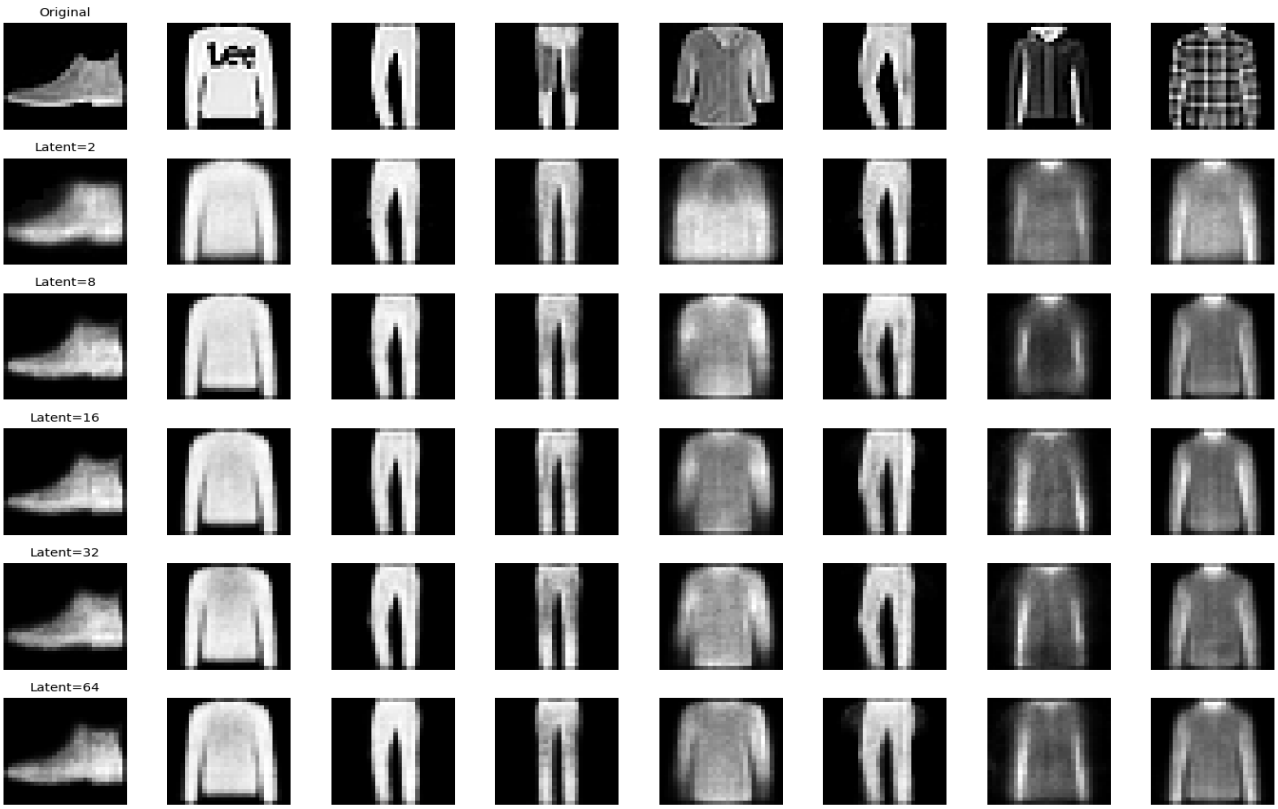

Figure 3. Originals and reconstructions of Fashion-MNIST images. Latent-2 model reconstructions are less detailed and blurry whereas bigger latent dimensions retain more visual structure.

The 2D plot in the latent space was also useful in structure. The autoencoder that was not trained to classify also learned loose clusters in the learned latent space, although this was not the intended goal. It means that even a reconstruction-based goal can generate useful interiorly depictions of the information (Hinton and Salakhutdinov, 2006; Vincent et al., 2010).

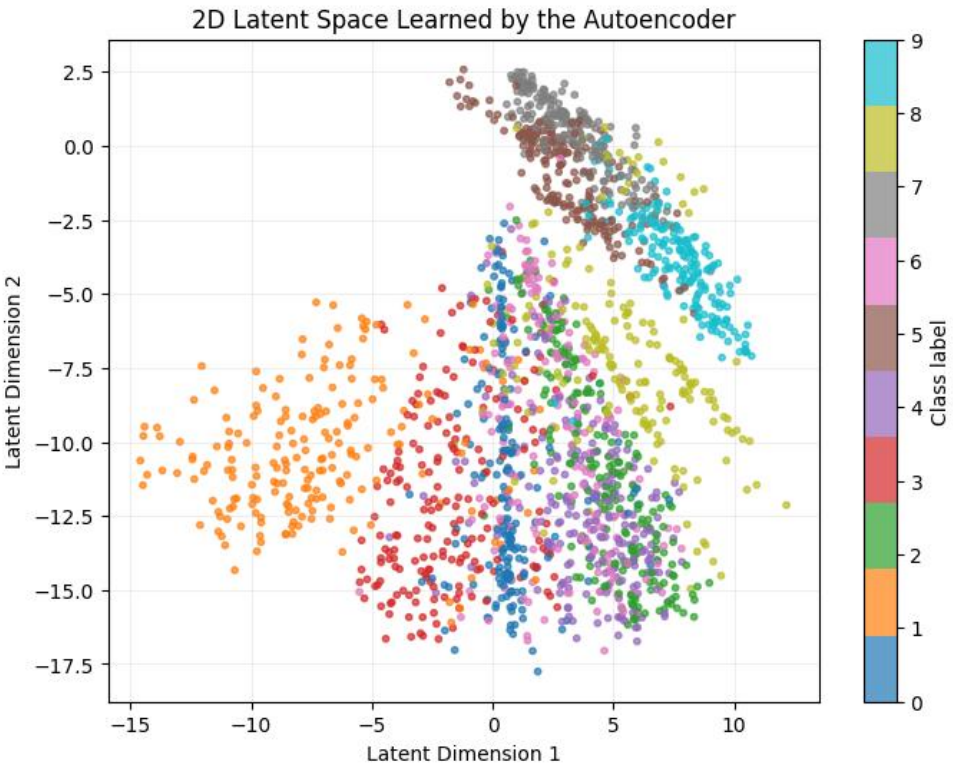

Figure 4. Two-dimensional latent space of autoencoder, where latent dimension is 2. A little loose clustering can be seen although the model is trained in the reconstruction mode.

Discussion

The success of the experiment is due to the fact that it clearly showed the trade-off between compression and reconstruction quality. The existence of a very small bottleneck causes the model to drop away excessively information, leading to increased reconstruction loss and blurrier outputs. Higher latent dimension is helpful in the reconstruction in the sense that the encoder is able to capture more meaningful detail. The gains however begin to level off when the latent space is large enough. This implies that the optimal latent dimension is not necessarily the largest dimension in the first place, but one that is both small in size and capable of approximating the reconstruction.

It is a valuable practical lesson to users of autoencoders. In practical use, the usefulness of the bottleneck size must be based on the objective. In the case the goal is a strong compression or low-dimensional visualisation, a smaller latent space could be beneficial, although quality of reconstruction could be reduced. A slightly bigger latent space could be more suitable in case the downstream task requires the preservation of features.

One of the weaknesses of this tutorial is that it employs a simple fully connected architecture. Convolutional autoencoders are commonly preferred in the case with image data since they are more effective at preserving local spatial patterns. The other weakness is that reconstruction loss is not sufficient to ensure that the learned representation is fair, robust or appropriate to apply in sensitive applications in the real world. In case the training data is biased or incomplete, the learnt latent space can also mirror such shortcomings (Goodfellow, Bengio and Courville, 2016).

The findings also indicate that there are diminishing returns: after the latent space has grown large enough to support the principal visual structure, further increases will only bring small gains.

Accessibility and teaching choices

A number of decisions were taken with the aim of making the tutorial available. The data is easily readable, the graph is not based only on colour, and the experiment is comparing few latent dimensions in a way that the trend is easily traced. Direct side by side pictures comparisons are also used in the tutorial and which are very effective to demonstrate what data is retained and what is lost.

Conclusion

This tutorial has explored the effectiveness of latent dimension in reconstruction fidelity in an autoencode that was trained on Fashion-MNIST. The results showed that latent dimension was a grave contributor of what will be remembered by the model. It was discovered that much smaller bottlenecks (2) were needed to squeeze in the data excessively, and moderate latent dimensions, especially 16 and 32, to store a lot of useful information and the reconstructions loss with a significantly larger proportion of the corruption.

The greater argument is that compression can only be positive when it is informative. The bottleneck well selection is thus the crucial consideration in using the autoencoders in practice Latent dimension is not a technical issue but rather a design decision, and influences directly the question of what the model would recall, what would it forget and what the learned representation would be useful.

References

Goodfellow, I., Bengio, Y. and Courville, A. (2016) *Deep Learning*. Cambridge, MA: MIT Press.

Hinton, G.E. and Salakhutdinov, R.R. (2006) ‘Reducing the dimensionality of data with neural networks’, *Science*, 313(5786), pp. 504–507.

PyTorch Documentation (n.d.) *PyTorch 2.x documentation*.

Vincent, P., Larochelle, H., Lajoie, I., Bengio, Y. and Manzagol, P.-A. (2010) ‘Stacked denoising autoencoders: Learning useful representations in a deep network with a local denoising criterion’, *Journal of Machine Learning Research*, 11, pp. 3371–3408.

Xiao, H., Rasul, K. and Vollgraf, R. (2017) *Fashion-MNIST: a Novel Image Dataset for Benchmarking Machine Learning Algorithms*. arXiv:1708.07747.